DAY 17 – Residual and Error Analysis

Day 17 focused on performing a deeper evaluation of the forecasting model by studying **residuals**—the difference between actual and predicted values. Residual analysis is crucial because it exposes hidden weaknesses in the model that basic metrics cannot reveal. Today's work emphasized understanding error patterns, diagnosing inconsistencies, and validating whether the model's assumptions hold true.

🔹 1. Understanding Residuals

Residuals help determine:

• How much the model deviates from real data
• Whether errors follow a pattern
• If the model is biased (predicts too high or too low)
• Where predictions fail during extreme market movement
• Whether the model generalizes well

A good model should have **small, randomly distributed residuals**.

🔸 2. Residual Plots and Their Importance

We analyzed several useful graphs:

✓ **Residual Line Plot** – Shows fluctuations in prediction error
✓ **Residual Histogram** – Displays frequency of error sizes
✓ **Residual vs Time Plot** – Reveals seasonality or trend-based errors
✓ **Q–Q Plot** – Checks if residuals follow a normal distribution

These visual tools help identify systematic issues.

🔸 3. Key Error Patterns Observed

Analysis revealed:

• Errors were higher during sudden price spikes
• Residuals were stable during normal trading days
• A slight bias appeared when prices rose rapidly
• Predictions were smoother than actual volatile data
• No major cyclical error pattern was found

These findings aligned with typical stock market forecasting challenges.

🔸 4. Diagnosing Model Weaknesses

Residual analysis helped us detect:

• Where the model overestimates or underestimates
• If predictions lag behind sharp movements
• How uncertainty affects sudden fluctuations
• Whether Prophet captured change points accurately

This ensures transparency before generating long-term forecasts.

🔸 5. Improving Model Using Residual Insights

Residual findings guide improvements like:

• Adding custom seasonality
• Adjusting changepoint sensitivity
• Revising data preprocessing
• Smoothing noisy regions
• Applying advanced tuning techniques

Residuals act as a foundation for refining overall model accuracy.

Key Takeaways from Day 17

• Residuals reveal hidden model errors
• Stable residual patterns indicate reliability
• Larger residuals appear during high volatility
• Residual plots help diagnose bias
• This analysis strengthens the forecasting process